If all 3,000 keep saving ₹350 a day, the agent pays for himself in year 3. Over 3 years, 78 loan referrals close the gap.

YOUR ASSUMPTION: EVERY SAVER KEEPS SAVING

Year 1

1,000

savers at year end

400 collections a day

every 2.5 days each

✓ no UPI needed

Year 2

2,000

savers at year end

800 collections a day

every 2.5 days each

! 50% must use UPI

Year 3

3,000

savers at year end

1,200 collections a day

every 2.5 days each

! 67% must use UPI

One person collects from about 400 savers a day (Arvinder's book is 200–400).

Every saver stays (your assumption)

SAVERS PUT IN ₹350 A DAY	YEAR 1	YEAR 2	YEAR 3	3 YEARS
Savers at year end	1,000	2,000	3,000	3,000
Saver-years (average saving)	542	1,542	2,542	4,625
Deposits (₹ crore)	4.55	12.95	21.35	38.85
Agent's ₹4 lakh a year, as % of deposits	0.88%	0.31%	0.19%	0.31%
Paid back: fund commission + other products	₹1,13,750	₹3,23,750	₹5,33,750	₹9,71,250
Gap left (+ = surplus)	**₹2,86,250**	**₹76,250**	**+₹1,33,750**	**₹2,28,750**
Referred ₹1.5L shop loans to fill it	**98**	**26**	**0**	**78**
… per saver a year (evidence 0.03)	0.181	0.017	0.000	0.017
Or own-book loans at 6% losses	62	17	0	50

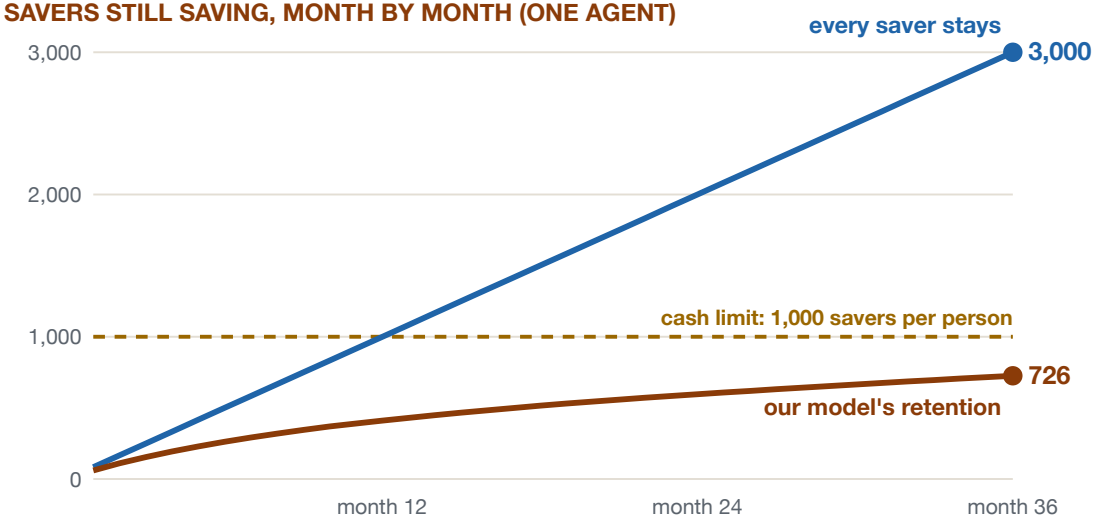
Year 3 earns a surplus, which pays back part of years 1–2; the 3-year column nets it.

It works only if two in three savers pay by UPI by year 3: one person cannot collect cash from 3,000 people.

Assumptions: agent ₹4,00,000 a year (your figure) · 1,000 sign-ups a year, joining at the start of each month, for 3 years (your figure) · collected every 2.5 days (your 2–3) · ₹350 a day per saver (middle of Arvinder's ₹200–500), 240 deposit days a year · one person collects from 400 savers a day (top of Arvinder's 200–400) · fund commission 0.35% a year on half a year's deposits (yearly terms) · other products ₹63 per saver a year (XO-39) · referral ₹2,925 per ₹1.5 lakh shop loan (2% less 2.5% taken back, XL-13) · own book ₹4,594 per loan at 6% losses (tables/lend_loan.csv) · retention: our model's curve at today's evidence, 24.2% still saving at 12 months, 14.5% at 24, 11.8% at 36 · evidence loans 0.03 per saver a year (XL-48). Built by model/agent_3yr.py → tables/agent3_*.csv.

With today's retention the agent has 726 savers after 3 years, not 3,000, and needs 306 loan referrals: 1 for every 5 savers a year.

SAVERS STILL SAVING, MONTH BY MONTH (ONE AGENT)



If savers put in ₹100 a day instead (3 years)

	AGENT COST	GAP	REFERRALS PER SAVER A YEAR	OWN-BOOK PER SAVER A YEAR
Every saver stays	1.08%	₹7,14,375	0.053	0.034
Our model's retention	3.45%	₹10,47,897	0.247	0.157

Our model's retention: 24 of 100 still saving after a year

SAVERS PUT IN ₹350 A DAY	YEAR 1	YEAR 2	YEAR 3	3 YEARS
Savers at year end	411	596	726	726
Saver-years (average saving)	261	519	668	1,449
Deposits (₹ crore)	2.19	4.36	5.61	12.17
Agent's ₹4 lakh a year, as % of deposits	1.82%	0.92%	0.71%	0.99%
Paid back: fund commission + other products	₹54,866	₹1,08,965	₹1,40,375	₹3,04,206
Gap left (+ = surplus)	**₹3,45,134**	**₹2,91,035**	**₹2,59,625**	**₹8,95,794**
Referred ₹1.5L shop loans to fill it	**118**	**99**	**89**	**306**
… per saver a year (evidence 0.03)	0.452	0.192	0.133	0.211
Or own-book loans at 6% losses	75	63	57	195

Evidence middle is 0.03 loans per saver a year, so year 3 alone needs 4× that; year 1 needs 15×.

Retention decides it, not the agent's cost: the same agent is 0.31% of deposits if savers stay, and short ₹8.96 lakh if they behave like today's daily savers.

Assumptions: agent ₹4,00,000 a year (your figure) · 1,000 sign-ups a year, joining at the start of each month, for 3 years (your figure) · collected every 2.5 days (your 2–3) · ₹350 a day per saver (middle of Arvinder's ₹200–500), 240 deposit days a year · one person collects from 400 savers a day (top of Arvinder's 200–400) · fund commission 0.35% a year on half a year's deposits (yearly terms) · other products ₹63 per saver a year (XO-39) · referral ₹2,925 per ₹1.5 lakh shop loan (2% less 2.5% taken back, XL-13) · own book ₹4,594 per loan at 6% losses (tables/lend_loan.csv) · retention: our model's curve at today's evidence, 24.2% still saving at 12 months, 14.5% at 24, 11.8% at 36 · evidence loans 0.03 per saver a year (XL-48). Built by model/agent_3yr.py → tables/agent3_*.csv.